



ȘTEFAN LUCIAN TABACU

Full Professor, PhD, Habilitation, Eng.

Automotive Engineering | Teaching and Research

PROFESSIONAL EXPERIENCE

HIGHER EDUCATION

Full Professor | 01.10.2016–present

National University of Science and Technology POLITEHNICA Bucharest, Pitești University Centre (formerly University of Pitești)
Teaching and research activities.

Junior Lecturer, Assistant Lecturer, Lecturer, Associate Professor | 01.10.1999–01.10.2016

University of Pitești
Teaching and research activities.

INDUSTRY AND TECHNICAL EXPERTISE

Court-Appointed Technical Expert | 01.05.2010–

Ministry of Justice
Judicial and extra-judicial technical assessments in road traffic and motor vehicles.

Engineer – computational analysis and product development | 01.10.2010–present

S.C. ALSECA ENGINEERING S.R.L.
Automotive components.

Engineer – computational analysis and product development | 01.04.2008–present

S.C. AKA AUTOMOTIV S.R.L., Călinești, Argeș County
Automotive components.

Research Scientist II | 01.10.2014–01.06.2015

S.C. ALSECA AUTOMOTIVE S.A., Tunari, Ilfov County
Computational analysis and product development.

Research Engineer | 01.09.2006–01.09.2008

S.C. MECDIN SISTEM S.R.L.
Computational analysis and services for the automotive and aerospace industries.

CONTACT INFORMATION

National University of Science and Technology POLITEHNICA Bucharest

Pitești University Centre · Faculty of Mechanics and Technology · Department of Vehicles and Transportation

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110040 Pitești, Argeș County, Romania
Nationality: Romanian

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ACADEMIC PROFILES

ORCID

[0000-0002-5736-4724](https://orcid.org/0000-0002-5736-4724)

ResearcherID

[A-4218-2009](https://www.researcherid.com/A-4218-2009)

Scopus Author ID

[24278948900](https://orcid.org/24278948900)

Publons

BrainMap

ResearchGate

COMPETENCIES

- Participation in scientific research programmes
- Design and fabrication of devices for specific applications
- Computer programming and use
- Computer-aided design
- Computational analysis

LANGUAGES

Romanian

Native language

English

B2 – listening, reading, spoken interaction, spoken production and writing

French

Proficiency level not specified

ADDITIONAL INFORMATION

Driving licence: Category B

Coordinator of the master's programme in Automotive Computer-Aided Engineering

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EDUCATION AND TEACHING

EDUCATION AND TRAINING

Habilitation – doctoral supervisor qualification | 2015

University of Pitești

Field: Automotive Engineering.

PhD | 1999–2002

University of Pitești

Engineering Sciences / Road Vehicles. Qualification level: Doctor / Engineering Sciences / Mechanical Engineering.

Diploma Engineer (Engineering degree) | 1994–1999

University of Pitești

TEACHING EXPERIENCE

Courses taught at bachelor's, master's and doctoral levels:

- Vehicle Bodies and Structural Analysis (bachelor's)
- Vehicle Dynamics (bachelor's and master's)
- Vehicle Design and Construction (bachelor's)
- Computer-Aided Design – CATIA (master's)
- Structural Analysis (master's and doctoral)

ACADEMIC RESPONSIBILITIES

Coordinator of the master's degree programme in Automotive Computer-Aided Engineering at the Faculty of Mechanics and Technology, University of Pitești.

Professional activities include teaching and research, computational analysis and product development for the automotive industry, and court-appointed technical expertise.

RESEARCH PROJECTS

Projects coordinated as project director

- “Structural solutions to optimise the load-bearing structure of Romanian-manufactured passenger cars” – CNCSIS AT120/2004.
- “Numerical models for the study of vehicle collisions” – CNCSIS AT107/2006.
- Study of the roll-over behavior of the passenger structure of the Roman 10215 – S.C. IRA Câmpina S.A., 2005.
- Analysis of mechanical systems – AKA Automotiv, 2007.
- VSR Clamp Life Study – Honeywell Garrett, 2008.
- VSR Balancing – Honeywell Garrett, 2010.
- Analysis of mechanical systems – S.C. GICNOSAG Metal S.A., 2011.

AREAS REFLECTED IN PROFESSIONAL ACTIVITY

- Automotive Engineering
- Computational analysis and product development
- Structural and impact analysis
- Auxetic structures and additive manufacturing

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SCIENTIFIC ACTIVITY AND PATENTS

SCIENTIFIC REVIEWER

Advances in Engineering Software; Aircraft Engineering and Aerospace Technology; Applied Mathematical Modelling; Applied Sciences; Engineering with Computers; IEEE Access; International Journal of Crashworthiness; International Journal of Mechanical Sciences; Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering; Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering; Journal of Advanced Manufacturing Technology; Journal of Applied Mechanics; Journal of the Brazilian Society of Mechanical Sciences and Engineering; Latin American Journal of Solids and Structures; Materials; Materials & Design; Mathematical Problems in Engineering; Mechanics Based Design of Structures and Machines; Physica Status Solidi B: Basic Solid State Physics; Thin-Walled Structures.

Member of the Editorial Board of the International Journal of Crashworthiness.

GRANTED PATENTS

RO129003-B1 – “Gradual door stop, comprises indexing rod provided at one end with hole for mounting rivet, and other end with buffer head in shape of T in horizontal plane.” Inventors: Cagri I., Stroe S., Tabacu S. Assignee: AKA Automotiv S.R.L. Derwent: 2013-W70567.

EP4155494-B1 – “Temporary clamping mechanism for maintaining e.g. car door, open in fixed position, has temporary stopper placed in plane perpendicularly to hinge shaft and perpendicular to mobile hinge element surface.” Inventors: Stirosu M., Tabacu S., Dinu N., Inal C. Assignee: AKA Automotiv S.R.L. Derwent: 2023-32059Q.

PATENT APPLICATIONS

RO130842-A0 – “Bearing panel for baggage compartment of motor vehicle which is provided with space for storing spare wheel.” Inventors: Diaconescu C.C., Leuciuc D., Oltean A., Tabacu S. Assignee: ALSECA Automotive S.A. Derwent: 2016-12077F.

RO129365-A2 – “Biodegradable composite material.” Inventors: Avadanei L., Badita M., Caramitu A.R., Groza C., Ion I., Mitrea S., Nicolae V., Sbarcea G., Tabacu I., Tabacu S. Assignees: INCD Electrochimie, University of Pitești and ICEFS COM S.R.L. Derwent: 2014-H82687.

PUBLICATION THEMES

- Numerical and experimental modelling
- Auxetic and anti-tetrachiral configurations
- Impact and crushing behaviour of automotive structures
- Components and structures manufactured by FDM and SLM
- Mechanical analysis of materials and structures

PERIOD

2008–2026

CV date: 10 February 2026.

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SELECTED PUBLICATIONS

- Badea, A.-G.; Tabacu, S.; Aparaschivei, A.-I.; Negrea, D.; Moga, S.; Ducu, C.** *Mechanical Characterization and Numerical Modeling of 316 Stainless Steel Specimens Fabricated Using SLM.* *Journal of Manufacturing and Materials Processing*, 10, 29 (2026).
- Stirosu, M.; Tabacu, S.; Cimpeanu, G.** *Investigation on the Manufacturing, Testing, and Simulation Processes of the Hood Hinge Assembly.* *Vehicles*, 7, 157 (2025).
- Tabacu, S.; Badea, A.-G.; Aparaschivei, A.-I.; Anghel, D.-C.** *Parametric Analysis of Auxetic Honeycombs.* *Mathematics*, 13, 1676 (2025).
- Tabacu, S.; Badea, A.; Sandu, A.** *Complex Analysis of an Auxetic Structure under Compressive Loads.* *Sustainability (Switzerland)*, 15(8), 6805 (2023).
- Tabacu, S.; Popa, D.** *Backward-Facing Analysis for the Preliminary Estimation of the Vehicle Fuel Consumption.* *Sustainability (Switzerland)*, 15(6), 5344 (2023).
- Tabacu, S.; Stănescu, N.D.** *Axial Compression and Buckling Analysis of Columnar Structures with Tetra-Anti-Chiral Configuration.* *Journal of Applied Mechanics – Transactions of ASME*, 89(8), 081007 (2022).
- Tabacu, S.; Stănescu, N.D.** *A theoretical model for the estimate of the reaction force for 3D auxetic anti-tetra chiral tubular structures under tensile loads.* *Thin-Walled Structures*, 168, 108304 (2021).
- Tabacu, S.; Predoiu, P.; Negrea, R.** *A theoretical model for the estimate of plateau force for the crushing process of 3D auxetic anti-tetra chiral structures.* *International Journal of Mechanical Sciences*, 199, 106405 (2021).
- Tabacu, S.; Negrea, R.F.; Negrea, D.** *Experimental, numerical and analytical investigation of 2D tetra-anti-chiral structure under compressive loads.* *Thin-Walled Structures*, 155, 106929 (2020).
- Tabacu, S.; Ducu, C.** *Numerical investigations of 3D printed structures under compressive loads using damage and fracture criterion: Experiments, parameter identification, and validation.* *Thin-Walled Structures*, vol. 39, art. 100775 (2020).
- Tabacu, S.; Ducu, C.** *An analytical solution for the estimate of the mean crushing force of structures with polygonal and star-shaped cross-sections subjected to axial load.* *International Journal of Mechanical Sciences*, 161–162, 105010 (2019).
- Tabacu, S.; Ducu, C.** *Experimental testing and numerical analysis of FDM multi-cell inserts and hybrid structures.* *Thin-Walled Structures*, 129, 197–212 (2018).
- Tabacu, S.** *Analysis of circular tubes with rectangular multi-cell insert under oblique impact loads.* *Thin-Walled Structures*, 106, 129–147 (2016).
- Tabacu, S.** *Axial crushing of circular structures with rectangular multi-cell insert.* *Thin-Walled Structures*, 95, 297–309 (2015).
- Tabacu, S.** *Numerical model (switchable/dual model) of the human head for rigid body and finite elements applications.* *Computer Methods in Biomechanics and Biomedical Engineering*, 18, 769–781 (2015).
- Tabacu, S.; Parlac, S.; Tomus, C.; Baba, V.** *Computational modelling of vehicle interior components for impact applications: moulding residual stress.* *International Journal of Crashworthiness*, 17, 591–605 (2012).
- Tabacu, S.; Tabacu, I.; Hadar, A.** *Computational modelling of vehicle interior components for impact application: Thickness analysis.* *International Journal of Crashworthiness*, 16, 421–438 (2011).
- Tabacu, S.; Pandrea, N.** *Numerical (analytical-based) model for the study of vehicle frontal collision.* *International Journal of Crashworthiness*, 13, 387–410 (2008).